Auteur: Victor Pena
Studierichting: Informatica
Oefeningenreeks 1C nr. 45.2

$$z^3 = -1 = -1 + 0i \Rightarrow \left\{ \begin{array}{l} x = -1 \\ y = 0 \end{array} \right.$$

$$r = \sqrt{(-1)^2 + 0^2} = 1$$

$$y = 0, x < 0 \Rightarrow \theta = \pi$$

$$\begin{aligned} w_0 &= \sqrt[3]{1}\,\mathrm{cis}\left(\frac{\pi + (0)2\pi}{3}\right) \\ &= \mathrm{cis}\left(\frac{\pi}{3}\right) \\ &= \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} \\ &= \frac{1}{2} + \frac{\sqrt{3}}{2}i \end{aligned}$$

$$\begin{aligned} w_1 &= \sqrt[3]{1}\,\mathrm{cis}\left(\frac{\pi + (1)2\pi}{3}\right) \\ &= \mathrm{cis}\,(\pi) \\ &= \cos\pi + i\sin\pi \\ &= -1 + 0i \\ &= -1 \end{aligned}$$

$$\begin{aligned} w_2 &= \sqrt[3]{1}\,\mathrm{cis}\left(\frac{\pi + (2)2\pi}{3}\right) \\ &= \mathrm{cis}\left(\frac{5\pi}{3}\right) \\ &= \cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3} \\ &= \frac{1}{2} - \frac{\sqrt{3}}{2}i \end{aligned}$$

References